Differential Equations and Slope Fields

ANSWER KEY

Verifying solutions to differential equations

1 Verify that $y = Ce^{3x}$ is a solution to the differential equation $\frac{dy}{dx} = 3y$ for any constant C.
$\frac{dy}{dx} = 3Ce^{3x} = 3(Ce^{3x}) = 3y$. This confirms $y = Ce^{3x}$ satisfies the equation for any C.

2 Verify that $y = x^2 + C$ is a general solution to $\frac{dy}{dx} = 2x$, and find the particular solution satisfying $y(1) = 5$.
$\frac{dy}{dx} = 2x$ ✓. Using $y(1) = 5$: $1 + C = 5 \Rightarrow C = 4$. Particular solution: $y = x^2 + 4$.

Solving separable differential equations

3 Solve the separable differential equation $\frac{dy}{dx} = \frac{x}{y}$, given $y(0) = 3$.
$y\,dy = x\,dx \Rightarrow \frac{y^2}{2} = \frac{x^2}{2} + C \Rightarrow y^2 = x^2 + 2C$. Using $y(0) = 3$: $9 = 0 + 2C \Rightarrow 2C = 9$. Solution: $y^2 = x^2 + 9$, i.e. $y = \sqrt{x^2 + 9}$ (positive root, since $y(0) = 3 > 0$).

4 Solve $\frac{dy}{dx} = 2xy$ given $y(0) = 1$.
$\frac{1}{y}dy = 2x\,dx \Rightarrow \ln|y| = x^2 + C$. Using $y(0) = 1$: $\ln 1 = 0 + C \Rightarrow C = 0$. So $\ln y = x^2 \Rightarrow y = e^{x^2}$.

5 Solve $\frac{dy}{dx} = \frac{\cos x}{y^2}$ given $y(0) = 2$.
$y^2\,dy = \cos x\,dx \Rightarrow \frac{y^3}{3} = \sin x + C$. Using $y(0) = 2$: $\frac{8}{3} = 0 + C \Rightarrow C = \frac{8}{3}$. So $y^3 = 3\sin x + 8$, i.e. $y = \sqrt[3]{3\sin x + 8}$.

Slope fields

6 For the differential equation $\frac{dy}{dx} = x - y$, find the slope at the points $(0,0)$, $(1,0)$, and $(1,1)$, as would be drawn on a slope field.

At $(0,0)$: slope $= 0 - 0 = 0$. At $(1,0)$: slope $= 1 - 0 = 1$. At $(1,1)$: slope $= 1 - 1 = 0$.

7 Explain how a slope field for $\frac{dy}{dx} = ky$ (exponential growth/decay) would look different for $k > 0$ versus $k < 0$.

For $k > 0$: slopes are positive above the x-axis and negative below, steepening as $|y|$ increases — curves grow away from $y = 0$ (unstable equilibrium, exponential growth). For $k < 0$: slopes point toward $y = 0$ from both sides, flattening as $|y| \to 0$ — curves decay toward $y = 0$ (stable equilibrium, exponential decay).

Exponential growth and decay models

8 A bacteria culture grows according to $\frac{dP}{dt} = 0.2P$, with $P(0) = 500$. Find $P(t)$ and the population after 10 hours.

Solution form: $P(t) = P_0 e^{kt} = 500e^{0.2t}$. $P(10) = 500e^2 \approx 500(7.389) \approx 3694$ bacteria.

9 A radioactive substance decays according to $\frac{dA}{dt} = -0.05A$. If $A(0) = 100$ grams, find how long it takes to decay to 50 grams (the half-life).

$A(t) = 100e^{-0.05t}$. Set $50 = 100e^{-0.05t}$: $0.5 = e^{-0.05t}$, $\ln(0.5) = -0.05t$, $t = \frac{\ln(0.5)}{-0.05} = \frac{-0.693}{-0.05} \approx 13.86$ years.

Matching slope fields to differential equations

10 A slope field shows horizontal segments (slope 0) along the entire x-axis and along the entire y-axis, with slopes elsewhere resembling the product xy. Identify the differential equation this slope field represents.

$\frac{dy}{dx} = xy$ — slope is 0 whenever $x = 0$ or $y = 0$, matching the described field.

Newton's Law of Cooling

11 An object cools according to $\frac{dT}{dt} = -k(T - 70)$, where $70°$F is room temperature. If $T(0) = 200°$F and $k = 0.1$, find $T(t)$.

Let $y = T - 70$: $\frac{dy}{dt} = -0.1y \Rightarrow y = y_0 e^{-0.1t}$. $y_0 = 200 - 70 = 130$. So $T(t) = 70 + 130e^{-0.1t}$.